Solution to Alice in Subtractionland

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(a) Alice wins whenever $k+1$ does not divide n, and Bob wins otherwise.

Our first observation is that this game is entirely symmetric: the outcome depends only on whom the first player is. Therefore, we can let a $\mathcal{P}$-position be a game state for which the current player wins, while an $\mathcal{N}$-position is a game state for which the next player to move wins. A crucial consequence of this is that if the game state is currently a $\mathcal{P}$-position, then it will be an $\mathcal{N}$-position on the next move, and vice versa.

Now, let S denote the starting state of the game, and consider the case where $n \leq k+1$ as an example.

1. When $n < k+1$, Alice can simply remove the entire pile to win. This means S is a $\mathcal{P}$-position.
2. When $n = k+1$, Alice's move removes up to k tokens, so there will be between 1 and k remaining tokens. We saw above that this is a $\mathcal{P}$-position, which means the previous game state, S, was an $\mathcal{N}$-position; hence, Bob wins.

We can inductively apply this logic to generalize. Suppose $n = q(k+1) + r$ for integers q, r with $0 \leq r < q$. Then, we induct on q. The base case where $q = 0$ was dealt with above, so we consider some arbitrary choice of q, with the claim that S is a $\mathcal{P}$-position if and only if $r \neq 0$. If this is indeed true for this choice of q, then consider what happens when q is incremented — that is, $k+1$ new tokens are added, so $n = (q+1)(k+1) + r$. We analyze $r = 0$ and $r \neq 0$ separately now:

1. If $r = 0$, Alice's move leaves $n' = q(k+1) + r'$ tokens behind, and r' cannot be 0 since this would require a removal of $k+1 > k$ tokens. Hence, Alice's move results in a $\mathcal{P}$-position, meaning the original position is an $\mathcal{N}$-position.
2. If $r > 0$, then Alice can remove r tokens. Then, this is the game state where $r = 0$, which is an $\mathcal{N}$-position, so S is a $\mathcal{P}$-position.

Thus, the truth of the claim for q implies the truth of the claim for $q+1$, so the principle of mathematical induction ensures that the initially claimed answer is indeed correct.

(b) First, we resolve the case where we have two piles of equal length. Here, Bob can adopt the simple strategy of "copy Alice's move," which always wins. Thus two piles of equal length is always an $\mathcal{N}$-position.

Now suppose one pile has a tokens while the other has $a+b$ tokens. This game then becomes equivalent to the one-pile case with $n = b$, with the caveat that a player can try "wasting a move" by removing tokens from the pile with a tokens. However, this is only favorable when the game is in an $\mathcal{N}$ state, at which point the other player can also "pass a move" by removing the same number of tokens from the other pile, reverting back to the same scenario.

Therefore, a player wins the two-pile variant if and only if they win the one-pile variant with b tokens.

(c) We describe a recursive algorithm to make the optimal move. Let the m bottom-most weights be w_1, w_2, $\ldots$, w_m, with m being the smallest possible number such that

$$w_1 + \cdots + w_m > k.$$

A player is winning if and only if they can force these m tokens to be the only remaining tokens during the other player's turn. Then, we see that winning the game is equivalent to winning the game without those m tokens, so follow this same procedure to make the optimal move after removing those tokens.

For the base case, when the total weight of the tokens is a positive number at most k, remove all the tokens. If the total weight is zero, remove any number of tokens. (In fact, the total weight being zero after this reduction is a lose condition, so it doesn't matter what you do.)

Also, the game's outcome is not uniquely determined by n and k; one such instance with variable results is $n = 6$ and $k = 2$. When the tokens are $1, 2, 2, 1$ in order, Bob wins, but when the tokens are $1, 1, 2, 2$ in order, Alice wins.